
S/4HANA Clean Core Modernization

A free, non-commercial community pilot for automated legacy ABAP refactoring via SAP BTP CAP (Node.js/TS) and Developer Extensibility (ABAP Cloud RAP).

01. Understanding the Clean Core Strategy

What is Clean Core? In simple terms, it means keeping the standard SAP ERP core code completely untouched. When custom code is mixed directly with standard code, S/4HANA system updates become complex, slow, and expensive.

The SAP **Clean Core** strategy resolves this by separating custom extensions from the standard ERP software. Built as a **free, non-commercial community learning project**, Clean-Core.io acts as an educational, Do-It-Yourself (DIY) transformation co-pilot. It helps developers refactor unreleased legacy structures (like direct database select statements or obsolete modifications) into modern, standard-compliant, cloud-ready extensions.

02. Choosing Your Path: RAP vs. CAP

To keep the core clean, standard SAP guidelines define two main upgrade-safe paths for developers:

TECHNICAL DIMENSION	IN-APP ABAP CLOUD (RAP)	SIDE-BY-SIDE (BTP CAP)
Core Target	SAP RAP runs directly inside the S/4HANA system.	SAP CAP runs externally on the SAP Business Technology Platform (BTP).
Database Access	Uses released CDS views and synchronous queries.	Decoupled using separate SQL databases and web APIs.
LUW Boundary	Runs natively inside standard database threads and locks.	Asynchronous, communicating via APIs or Event Mesh.
UI Layout	Built with standard SAP Fiori Elements layout templates.	Highly flexible (React, Angular, or customized web pages).
Best Fit Case	Immediate database updates, validations, and document locks.	Decoupled customer portals, SaaS, or unpredictable traffic scales.

Factual Architectural Rule: If the legacy logic needs to check data instantly and block standard SAP database threads during transaction saves, it belongs in the ABAP Cloud (RAP) track. If the process is asynchronous or integrates with non-SAP systems, it belongs in the SAP BTP (CAP) track.

03. Decoupling the Database Core

Direct select queries on unreleased legacy database tables (like `KNA1` for customer master data or `BSEG` for accounting data) cause upgrade compilation blocks. Decoupling means replacing raw queries with standard released SAP APIs.

The Clean-Core.io transformation pipeline automatically identifies unreleased legacy dependencies in your old ABAP code and maps them to target released standard scenario interfaces:

RELEASED SAP OBJECT MAPPING

Direct raw database tables are replaced with safe, standard released interfaces:

- ✓ `KNA1` (Customer) → `I_Customer` (CDS View)
- ✓ `BSEG` (Accounting) → `I_JournalEntry` (CDS View)
- ✓ `MARA` (Material) → `API_PRODUCT_SRV` (OData)
- ✓ `VBAK` (Sales Header) → `I_SalesOrderTP` (RAP)

ASYNCHRONOUS EVENT PUB/SUB

For BTP CAP side-by-side extensions, the platform generates a clean ABAP Cloud class (or BAdI trigger implementation) on the ERP side.

This class intercepts database changes in S/4HANA and publishes the event payload asynchronously to the **SAP Event Mesh**. The external Node.js microservice receives the message and processes it, achieving zero processing coupling.

WHY STANDARD APIS KEEP THE CORE CLEAN

By routing all requests through official, released APIs, SAP can modify its underlying database tables during upgrades without breaking your custom applications. The API acts as a stable contract between the standard ERP and your custom code.

04. Quality Engineering & Verification Sandboxes

Every modern software project requires rigorous testing. Clean-Core.io automatically equips your transformed cloud application with automated testing sandboxes to verify correct behavior:

PLAYWRIGHT & TAP TESTING SANDBOX (BTP CAP)

The system auto-generates express unit tests and runs them in a sandboxed, server-side Node.js environment.

By providing a **BTP Destination JSON**, you can securely test live connections to your non-productive S/4HANA tenant. The test runs through the **SAP Cloud Connector** (utilizing standard Principal Propagation tunnels to forward user context safely).

ABAP UNIT TEST DOUBLES (ABAP CLOUD RAP)

For in-app RAP structures, the platform generates a comprehensive ABAP Unit test suite. It uses standard local test doubles to mock database reads and writes. This allows verifying validation classes locally without modifying actual database tables.

SAP SIGNAVIO & SAP BUILD BPMN EXPORTS

The platform maps the L3 business process into a standard BPMN 2.0 XML schema. It defines explicit swimlanes and role-based tasks (e.g. Finance Analyst, Developer, CISO). The resulting XML file can be directly imported into **SAP Signavio Process Manager** or **SAP Build Process Automation** to visualize your target process.

05. Enterprise Governance & Data Safety

Proprietary legacy source code is a highly confidential business asset. Clean-Core.io is engineered with strict cryptographic and operational safety boundaries to ensure data compliance.

GDPR & EU SOVEREIGNTY INFRASTRUCTURE (ART. 25 GDPR)

All storage and compute layers are hosted entirely within the **Belgium (europe-west1)** GCP region, satisfying European data residency policies.

Art. 17 GDPR (Right to Erasure): Triggering account deletion initiates an atomic backend routine that permanently wipes Firebase Auth records, Firestore project documents, and stored analysis artifacts instantly.

STATELESS PROCESSING & BYOK VAULT

Bring Your Own Key (BYOK): Your private Gemini API key is encrypted directly inside your browser using AES-GCM 256-bit encryption (PBKDF2 with 100,000 iterations). Plaintext keys are never stored on our servers.

Stateless Vertex AI Proxying: All prompts are routed using enterprise-grade stateless configurations. In accordance with Vertex Cloud terms, code is strictly excluded from LLM reinforcement training loops. Your intellectual property remains strictly yours.

ZERO MASTER DATA PROCESSING

To prevent data leaks, Clean-Core.io only parses code structures and abstract schemas. It does not parse or process master data or actual business records (like customer names or pricing values).

06. Community Vision, Closed Pilot & Educational Design

Clean-Core.io is **strictly a free, non-commercial educational project** built for the SAP developer community. Its purpose is to demonstrate how modern AI-driven architectures can help automate legacy ABAP transitions and make Clean Core guidelines accessible and practical.

CLOSED PILOT ACCESS

The platform is accessible via a **free, restricted Closed Pilot**. New registrants receive 5 free transformations to explore the wizard. For unlimited, private, and fully secure enterprise modernizations, the platform supports **Bring Your Own Key (BYOK)**.

EDUCATIONAL DESIGN OVERLAYS

The app features contextual **glossary overlays** explaining complex security concepts (helmet.js, stateless JWT, principal propagation) and **deep-dive connectivity drawers** displaying ready-to-use Node.js client integrations.

GET STARTED WITH CLEAN CORE

Accelerate your S/4HANA modernization journey today. Upload legacy ABAP, map standard APIs, and bundle deployable CAP/RAP packages instantly.

LAUNCH MODERNIZATION WIZARD → [HTTPS://CLEAN-CORE.IO](https://clean-core.io)

Felix Frenzel
Clean-Core.io Community

Clean Core Council
Bamberg, Germany